

S3 + CloudFront + Route 53

The exact console steps, in the order they were actually done — with the reasoning behind each one.

1 Create the S3 bucket

- 01 S3 Console → **Create bucket**
- 02 Bucket type: **General purpose** · Namespace: **Global**
- 03 Bucket name: **milk-and-bean-site** (must be globally unique)
- 04 Region: eu-north-1, same as EC2
- 05 Leave **Block all public access** checked for now → Create bucket

WHY THIS STEP

S3 stores the static files without an always-on machine underneath — that absence of a running server is exactly what makes it cheaper than EC2.

2 Upload files & enable static hosting

- 01 Objects tab → Upload → index.html, style.css, img/
- 02 Properties tab → **Static website hosting** → Edit → Enable
- 03 Hosting type: **Host a static website** · Index document: **index.html**
- 04 Save — note the generated **website endpoint** URL

WHY THIS STEP

The website endpoint (not the plain bucket URL) knows to serve index.html at the root. This is the address CloudFront needs later — the plain bucket endpoint won't work.

3 Open public read access

- 01 Permissions tab → Block public access → Edit → uncheck all four → confirm
- 02 Same tab → **Bucket policy** → Edit → paste public-read JSON policy
- 03 Principal: * · Action: **s3:GetObject** · Resource: bucket ARN + /*

WHY THIS STEP

Removing Block Public Access only takes off the lock. The bucket policy is what actually opens the door — skip it, and every visitor gets Access Denied.

4 Create the CloudFront distribution

- 01 CloudFront Console → **Create distribution**
- 02 Distribution type: **Single website or app**
- 03 Origin: Browse S3 → select bucket → click **Use website endpoint**
- 04 Leave remaining settings default → Create → wait for **Deployed** (5–15 min)

WHY THIS STEP

CloudFront caches the site at edge locations near visitors (fixes 'slow abroad') and is the only piece in this stack that can hold a custom domain and free HTTPS certificate.

NOTE

Origin must be the S3 **website endpoint** (s3-website...), not the plain bucket endpoint — the plain one doesn't know to serve index.html automatically.

5 Buy a domain

- 01 Registered **milkandbean.store** via Namecheap

WHY THIS STEP

A real domain is required to use Route 53's Alias record with CloudFront — DuckDNS only supports A records (IP), not the CNAME/Alias pointing CloudFront needs.

6 Request a TLS certificate (ACM)

- 01 ACM Console → switch region to **US East (N. Virginia)** — required
- 02 Request a public certificate → domain: **milkandbean.store**
- 03 Validation method: **DNS validation** → Request
- 04 On the certificate page → Domains section → click **Create records in Route 53**
- 05 Wait for status: Pending validation → **Issued**

WHY THIS STEP

ACM proves domain ownership via a unique CNAME only the real owner could add. Skipping the 'Create records in Route 53' button leaves nothing for ACM to check — it will sit pending forever, not because it's slow, but because it has nothing to validate against.

7 Create the Route 53 hosted zone

- 01 Route 53 → Hosted zones → **Create hosted zone**
- 02 Domain: milkandbean.store · Type: **Public hosted zone**
- 03 Create (\$0.50/month) → copy all **4 nameserver values**

WHY THIS STEP

This is AWS's DNS record book for the domain. It has to exist before anything can be validated or aliased against it.

8 Point Namecheap at Route 53

- 01 Namecheap → domain → Domain tab → Nameservers
- 02 Switch to **Custom DNS** → paste all 4 Route 53 nameservers → Save

WHY THIS STEP

This is the actual handoff: it tells the internet to stop asking Namecheap where the domain points, and ask Route 53 instead. Can take minutes to hours to propagate.

9 Attach domain + certificate to CloudFront

- 01 Distribution → General tab → Edit
- 02 Alternate domain name (CNAME): **milkandbean.store**
- 03 Custom SSL certificate: select the issued ACM certificate
- 04 Save → wait for **Deployed**

WHY THIS STEP

Route 53's Alias picker only lists distributions that already have the target domain registered here. Skip this, and the distribution won't appear as an option in the next step.

10 Create the Route 53 Alias record

- 01 Route 53 → hosted zone → **Create record**
- 02 Name: blank (root domain) · Type: **A** · Alias: **On**
- 03 Route traffic to: **Alias to CloudFront distribution** → select it
- 04 Routing policy: Simple → Create records

WHY THIS STEP

The final wire — the record that actually resolves milkandbean.store to the CloudFront distribution when someone visits.

11 Test

- 01 Visit **https://milkandbean.store**
- 02 Confirm it loads over HTTPS on the custom domain

milkandbean.store → Route 53 Alias → CloudFront (HTTPS, edge-cached) → S3 (file storage)